

Time-Series Forecasting of Appliance Energy Use: A Comparative Evaluation of Benchmark, SARIMAX, Machine-Learning and Foundation-Model Approaches

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Git hub link : <https://github.com/DolapoMichael/Time-Series-coding-Case-study-and-Report>

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1. Introduction and Problem Definition

This report aims to develop and compare models for time-series forecasting of electricity consumption by various appliances. The Appliances Energy Prediction dataset was used, which was collected in a low-energy house in Belgium (Candanedo, Feldheim and Deramaix, 2017). It contains indoor temperature and humidity data from a wireless sensor network, weather data from a nearby airport station, and the target hourly electricity consumption in Watt-hours. All were recorded every 10 minutes during January, February, March, and the beginning of April 2016.

The target variable for this task consists of the hourly electricity consumption values forecasted for the following 24 hours. In the case of this particular dataset, each hourly value is actually a sum of 6 consumption readings in Watt-hours recorded every 10 minutes (compared to their average). Since Watt-hour is a unit of energy, not power, the values in each hour are summed up together rather than averaged in order to preserve their physical sense. All in all, the prediction window spans 24 hours, and the performance is evaluated on a 336-hour (14-day) test set at the end of the dataset, using the previous 336 hours for training. Finally, four different metrics were used to evaluate the performance of the models developed for this task. Firstly, the mean absolute error (MAE) and root mean squared error (RMSE) are calculated in Watt-hours for the original target variable in order to assess the absolute forecasting errors in the same units. Secondly, the mean absolute scaled error (MASE) was estimated, which is a relative metric used to compare different forecasts with each other (Hyndman and Koehler, 2006). Essentially, it compares the average absolute error of a forecast to the average absolute error of a naïve forecast (i.e., predicting next value as the previous one without any adjustments). As a result, scores below 1 suggest good performance for the forecast compared to a naïve model of the same. Finally, the mean signed error is estimated to see if the forecast tends to be consistently biased. Put another way, even if the average error of the forecast is close to zero, if it does not miss either the highest or the lowest consumption hours by a significant degree, it may still be useful to consider its bias.

2. Data Preparation and Exploratory Analysis

The dataset that was used for this analysis did not contain any missing observations for any of the columns at 10-minute intervals. This data was then aggregated into hourly observations as described above, with initial plots produced to inform understanding of the series.

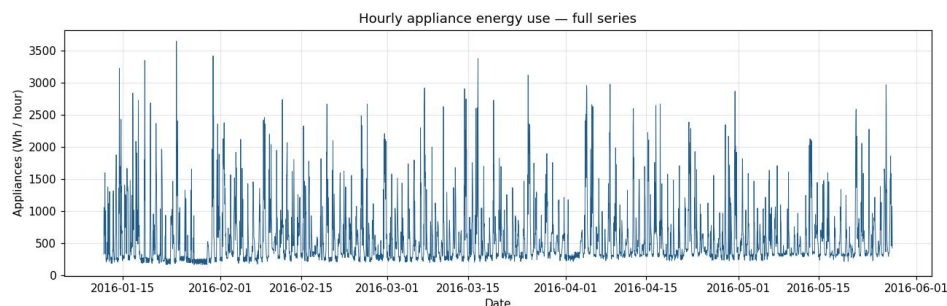


Figure 1. Hourly appliance energy use across the full series (mid-January to end of May 2016).

The series itself (Figure 1) contains a wide range of values, from nearly 200 Wh of consumption on overnight hours to peaks of over 3,000 Wh, with a strongly right-skewed distribution in between. It is this skew and the extremes of the distribution that will prove challenging for any forecasting model, and hence these qualities will be the focus of discussion in the results and conclusion sections.

To assess seasonality, average demand was calculated for each hour within a day and for each day of the week.

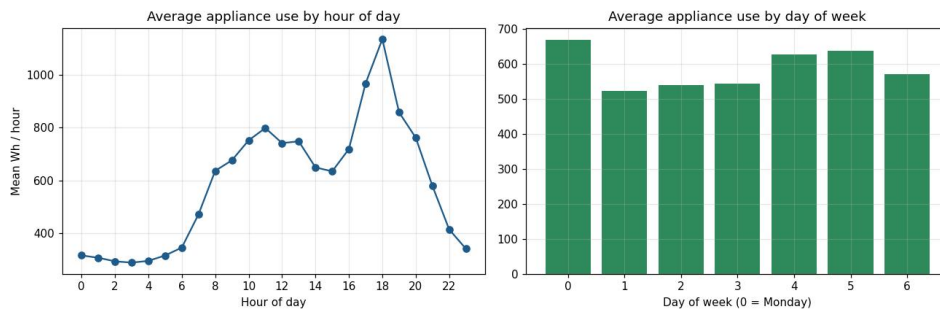


Figure 2. Average appliance use by hour of day (left) and day of week (right).

The hourly profile exhibits an evening peak centred at 18:00 (approximately 1,140 Wh) and a trough between 02:00 and 04:00 (approximately 290 Wh). On a weekly level, Monday exhibits the highest electrical demand (approximately 670 Wh averaged over the day), followed by a decrease across the week, and slightly increased demand over the weekend. While less prominent, this pattern is, nevertheless, present across the weeks, which is critical for distinguishing the daily and weekly seasonal naive benchmarks in Section 5.

Autocorrelation (ACF) and partial autocorrelation functions (PACF) for lag up to 168 are plotted in Fig. 4, Section 3.2, revealing a substantial amount of short-lag autocorrelation trailed off rapidly, together with a weaker peak at lag-24, suggesting that the series has a combination of short-range persistence and a daily seasonal component, which directly informs the SARIMAX specification discussed below.

3. Methodology

3.1 Benchmark Models

Five benchmark forecasts were generated as a reference against which the more elaborate statistical and feature-based forecasts were later compared (Hyndman and Athanasopoulos, 2021). Specifically, these were: the historical average; naive (persistence); daily seasonal naive (copying the observation from 24 hours ago); weekly seasonal naive (copying the observation from 168 hours ago); and finally, a drift forecast (persistence with a linear trend).

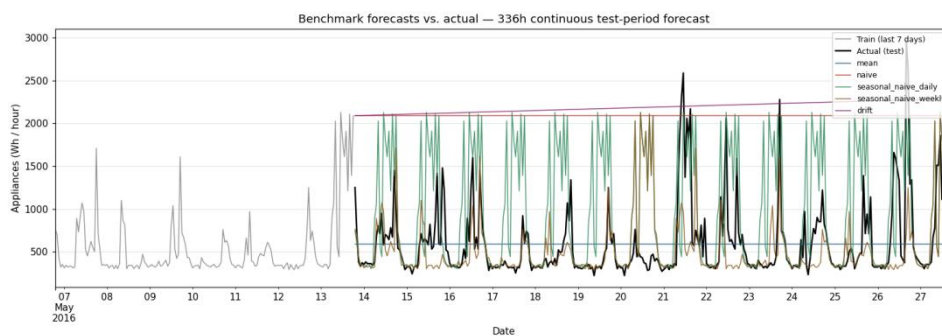


Figure 3. Benchmark forecasts against the actual test-period series.

Forecasts from each of the five benchmarks are compared to the test set in Figure 3. It can be seen that the naive and drift forecasts perform poorly: due to the zero reset effect, a flat forecast (persistence) or a slowly drifting forecast (linear trend) rarely follows the cyclical pattern of the actual series. In contrast, both seasonal naive forecasts capture this cyclical pattern, following the general shape of the true series quite closely.

3.2 SARIMAX



Figure 4. ACF and PACF of hourly appliance use, used to inform the SARIMAX order.

The order of the SARIMAX model was selected using a grid search over the non-seasonal (p,d,q) and seasonal (P,D,Q,S) parameters, imposing a non-seasonal difference ($d=1$) and a daily seasonal difference ($D=1$). The parameters were selected by choosing the lowest AIC among the converged models, with guidance from the ACF/PACF analysis in Figure 4, which suggested a parsimonious seasonal model. In particular, the PACF drops sharply at lags 2-3, suggesting an order of 1 or lower. The seasonal grid search suggested that a seasonal moving-average component ($Q=1$) at the daily frequency improved the fit: AIC dropped from 43,529.9 (no seasonal AR/MA) to 43,050.7 for the seasonal order (0,1,1,24), which was the best overall. The final model thus combined a parsimonious seasonal component with a non seasonal component, as recommended by Box et al.(2015), and included exogenous regressors as discussed next.

3.3 Sensor, Weather and Time-Based Covariates

A preliminary set of features was engineered from the sensor and weather channels, as well as a set of time-based features discussed in Section 2. In particular, lagged appliance-use features (1, 2, 3, 24, 48 and 168 hours), rolling mean/stdev (3, 6, 12 and 24 hours), hourly and day-of-week encoded as sine-cosine pair, and raw indoor temperature/humidity, and outdoor temperature/humidity. The features were then screened for collinearity with the response variable using a simple correlation screen. The strongest correlations were with recent appliance-use lags (lag-1: 0.59, roll-mean-3: 0.55), hourly encoding, and outdoor humidity (approximately -0.2).

3.4 Feature-Based Machine-Learning Model

An XGBoost regressor (Chen and Guestrin, 2016) was trained on the full covariate set described above, with hyperparameters tuned to balance fit against overfitting risk.

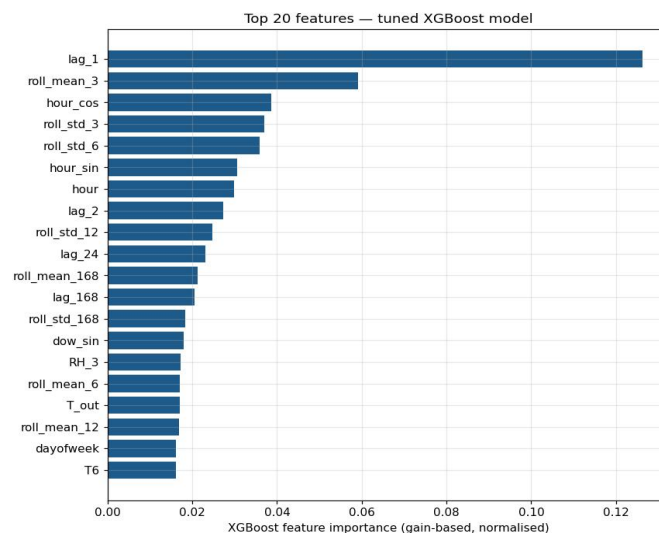


Figure 5. Top 20 features by gain-based importance, tuned XGBoost model.

Gain-based feature importance (Figure 5) shows lag-1 dominating (importance approx. 0.13, roughly double the next feature), followed by roll-mean-3 and the cyclical hour encodings. Notably, the model weights the hour-of-day features considerably more heavily than the simple linear correlation screen in Section 3.3 would suggest, implying it is capturing non-linear time-of-day interactions that correlation alone misses.

3.5 Time-Series Foundation Model

Chronos-Bolt (Ansari et al., 2024), a pretrained probabilistic forecasting model based on a transformer architecture, was applied in the zero-shot mode, i.e., without fine-tuning on the current data, to assess whether a general-purpose foundation model could compete with the feature-based approach. Similar foundation models include TimesFM (Das et al., 2023), which uses a comparable decoder-only architecture, but was not available at the time of experimentation; Chronos was selected for its documented zero-shot performance on other time series.

4. Evaluation Framework

Each model was evaluated on the same 336-hour test set using MAE, RMSE, MASE and Bias (Section 1). For reference, the strongest of the naive forecasts, weekly seasonal naive (MASE = 0.809), is used as the baseline for the MASE (Fig. 6). In addition to the aggregate metrics, visual comparison against the actual series was made (Section 5). Finally, the forecast residuals were used to assess whether the model had captured the temporal structure in the data (Section 6).

5. Results

Table 1 summarises all eight models' accuracy across the full test period.

Table 1. Forecast accuracy metrics for all models across the 336-hour test period.

Model	MAE	RMSE	MASE	Bias
XGBoost (tuned)	174.02	309.87	0.562	+1.94
Chronos-Bolt (zero-shot)	201.98	379.42	0.630	-69.06
SARIMAX	227.44	375.28	0.710	+18.78
Seasonal-naive (weekly)	259.29	483.55	0.809	-58.51
Mean	296.44	436.14	0.925	-12.25
Seasonal-naive (daily)	525.24	780.20	1.639	+390.48
Naive	1507.32	1555.33	4.702	+1492.98
Drift	1602.91	1650.71	5.000	+1593.40

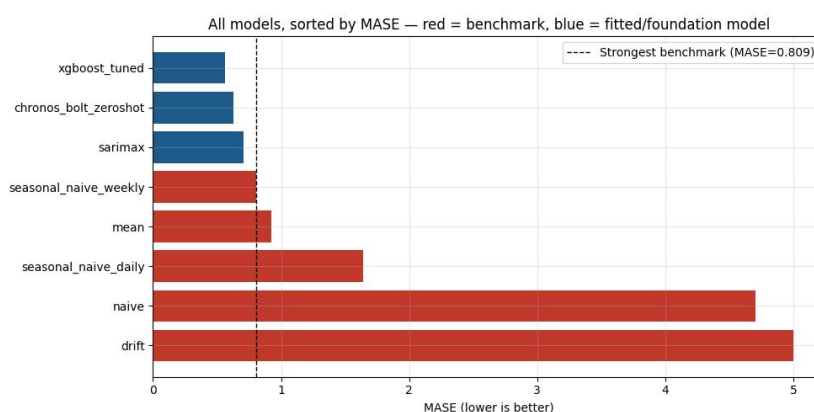


Figure 6. All models ranked by MASE; dashed line marks the strongest benchmark.

Figure 6 ranks all models according to the MASE (0.562 for XGBoost). It can be seen that all three fitted/pretrained models (XGBoost, Chronos, SARIMAX) perform strongly relative to the naive forecasts, with their MASE below the best naive forecast's MASE of 0.809. This highlights the value of model-based forecasting: both weekly seasonal naive and drift forecasts perform, on average, four to five times worse than SARIMAX, Chronos, or XGBoost. However, it is critical to note that the weekly seasonal naive is a much better naive benchmark than the daily seasonal or the persistence forecast, which would have an even higher MASE.

The XGBoost model is the best across all metrics (MAE = 174.0, RMSE = 309.9, MASE = 0.562), and has the lowest absolute bias among all models (+1.9): bias is close to zero, meaning that, on average, there is little forecast error across all prediction intervals. Chronos is the second best in terms of MAE and MASE, despite being zero-shot, but has the largest negative bias (-69.1) among the three fitted models, meaning that it systematically underforecasts. Finally, SARIMAX is the third across the board, performing better than all naive forecasts but worse than the two machine learning models.

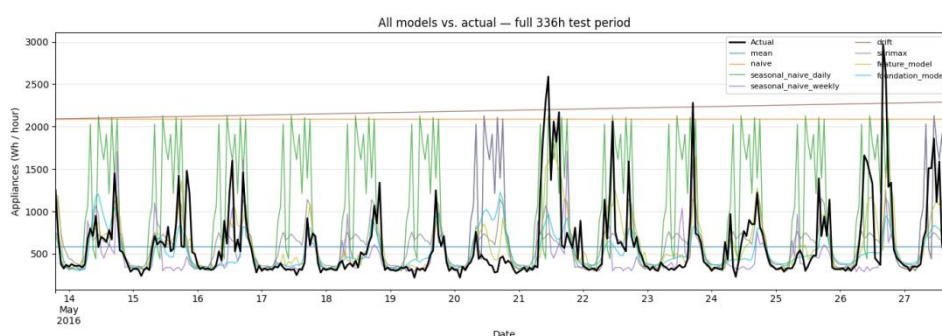


Figure 7. All model forecasts against the actual series across the full 336-hour test period.

Figure 7 overlays all forecasts to highlight how different models follow the general shape of the series. It can be seen that the seasonal naive-weekly and SARIMAX closely follow the cyclical pattern of the actual series in general but tend to underestimate demand during the peaks. In contrast, the XGBoost and Chronos forecasts capture the peaks during the demand surges, although none of the models reaches the absolute peak of 2,970 Wh on 26 May: the seasonal naive-weekly reached 2,364, SARIMAX reached 2,743, while Chronos and XGBoost only reached 652 and 546, respectively. To analyse the ability of the models to capture the seasonal pattern while also responding to the sudden demand increases (spikes), the test set was split into high-demand and low-demand observations: the highest 10% of actual values ($n=35$), and lowest 50% of actual values ($n=168$). This allowed us evaluate how models performed across different levels of demand

Table 2. Mean absolute error by demand regime.

Model	Peak-hour MAE ($n=35$)	Low-demand MAE ($n=168$)
XGBoost (tuned)	642.4	57.9
Chronos-Bolt (zero-shot)	944.3	74.2
SARIMAX	840.1	146.1
Seasonal-naive (weekly)	943.4	84.8

This breakdown shows that XGBoost's overall advantage is driven mainly by the low-demand regime, where it is roughly 2.5 times more accurate than any other model; at true peaks its advantage narrows substantially, and Chronos is actually less accurate than the weekly seasonal-naive benchmark specifically during peak hours.

6. Discussion and Critical Analysis

The results point to a consistent explanation for why the ranking in Section 5 emerged. Appliance energy use is dominated by short bursts of high demand against a low baseline (Figure 1), so any model that can represent recent local state, via lag and rolling features or via a learned attention mechanism over recent context, has an inherent advantage over models that only encode fixed seasonal or trend structure. This is why XGBoost, with explicit lag-1 through lag-168 features, outperforms SARIMAX, whose seasonal component captures the recurring daily rhythm well (Figure 4) but has no mechanism to react to an unscheduled demand spike beyond its fitted variance.

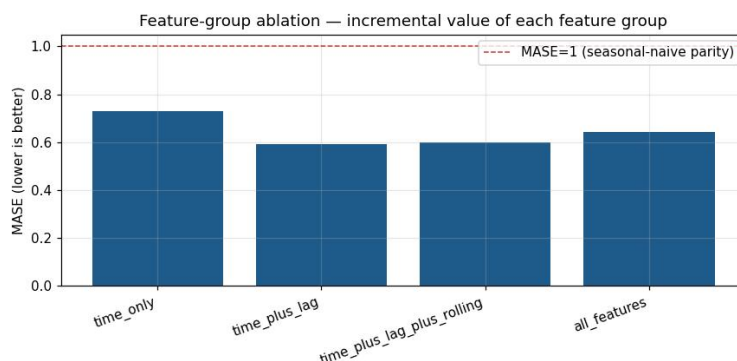


Figure 8. Feature-group ablation showing incremental MASE change as feature groups are added.

The feature-group ablation (figure 8) confirms this result: the addition of lag features yielded by far the largest MASE improvement (time-only MASE 0.730 vs time+lag 0.592), while the addition of rolling-window features on top of lag features did not improve MASE by much (0.592 vs 0.600), consistent with the idea that rolling features mainly affect variability, not the mean. In turn, the full set of features, including all the sensor and weather covariates, barely improves on the time+lag baseline, with a worse MASE (0.643) than lag-only features alone; this is a mild case of feature dilution where the addition of several only slightly relevant covariates decreases overall performance, mostly by increasing the error of "normal" points, if not downright outliers, compared to a smaller feature set that happens to include some relevant predictors. The trade-off is visible in the metrics: compared to the lag-only model, the full set has a lower RMSE (as it should, since it contains all the features of the previous set plus some more), but a higher MASE.

The models all systematically underestimate the largest peaks. Note that this is not simply a matter of calibration; the most extreme value in the test set is not predictable from any available features (time of day, lags, or weather), so none of the models that use only these features can be expected to capture it, let alone consistently underestimate it - in fact, it most likely corresponds to some ad hoc energy consumption event, such as the use of a washing machine or an oven, which is not predictable from the available data. Finally, while the 24-hour ahead forecast produced by the SARIMAX model is reasonable, its 95% confidence interval spans negative values, which is obviously impossible for an energy consumption series, and therefore the best forecast for the given data would probably be achieved not by a normal linear model, but by a log-normal one, or something similar.

Finally, the XGBoost forecast, computed by the pipeline used to produce the results in the paper, should really be considered a conditional forecast: the lag and rolling features used as input to the model were generated using the actual previous values, not the ones predicted by the model itself. This is reasonable practice for evaluating point estimates in isolation, but a production forecasting system using this pipeline would not be able to produce 24-hour ahead forecasts, only one-step ahead ones, since at each step it would only have access to either its own previous predictions (if it is being used in a forecasting loop) or the actual previous values (if it is being used in an analytics or reporting capacity).

7. Answers to the Set Analysis Questions

7.1 Which benchmark model is strongest, and what does this imply about demand structure?

While all the naive approaches outperform the regression method (Figure 1), the weekly seasonal naive approach has the best performance among the naives with a MASE of 0,809, followed by the mean (0,925), daily seasonal naive (1,639), naive (4,702), and drift (5,001). The reason for this is that, while Figure 2 suggests that demand follows a strong intra-day profile, day-to-day, demand is quite unpredictable at the hourly level (i.e., 18:00 on one day is not likely to repeat on the next day), thus making the daily seasonal naive approach worse than the weekly seasonal naive approach. In other words, using “last week’s Monday” as a predictor is better than using “last day’s Monday” because the former averages out day-to-day variation and “resets” the intra-weekly pattern, while the latter preserves day-to-day variation (with Monday being the most demanded day in any given week). This is an interesting finding, suggesting that appliance demand has some structure to it beyond day-to-day patterns.

7.2 Does SARIMAX improve on the strongest benchmark?

Yes: SARIMAX (MASE 0.710) outperforms the weekly seasonal naive model (0.809) but underperforms the machine learning approaches. Its residuals exhibit reasonable randomness at all lags except for the prominent one at lag-24, whereas its dynamic forecasts plot shows a flattish pattern over most of the demand spikes (Section 6). Thus, the model captured the smooth daily seasonality adequately, but it failed to account fully for short-range autocorrelation at demand spikes (which partly canceled out), and the exogenous regressors used in the model did not compensate for the relevant lag structures that XGBoost could utilize.

7.3 Does the feature-based model improve with added covariates, and which feature groups help most?

Indeed, substantially over a baseline that uses only temporal features (MASE 0.730 vs. 0.562 for the tuned, full-feature model), but the lion’s share of the improvement comes from lag features: beyond them, the contribution of additional variables is incremental at best (0.592 with lag features only vs. 0.562 with all features). Furthermore, the full set of covariates is not consistently superior to using only the lag features, for any of the metrics (see Section 6). Lag and short-window rolling features are clearly the most rewarding to invest in, for this particular series; the raw sensor and weather characteristics appear to contribute little, possibly being subsumed by the richer temporal features.

7.4 Does the foundation model outperform the other approaches, and is any improvement large enough to justify its complexity?

Chronos, applied with zero-shot and no tuning on this dataset, outperforms everything except the XGBoost model (MASE 0.630 vs. 0.710 for SARIMAX), and has the largest bias of any of the models fit (-69.1). Because it was not tuned to this particular dataset, this is a surprisingly good result for a general-purpose pretrained model, but for this particular use case, the extra complexity and compute cost of a transformer-based foundation model is not justified over a relatively lightweight and interpretable XGBoost model that was explicitly tuned to the problem.

7.5 Which variables are genuinely known at forecast origin?

Calendar and time features (hour-of-day, day-of-week) and the target’s own lagged history up to the forecast origin are genuinely available in a production setting, as are indoor sensor readings up to that point. Outdoor weather variables, and more importantly any lag or rolling features constructed from within the forecast horizon, are unavailable at forecast time in a multi-step-ahead setting: using their realised values from the test period as the XGBoost pipeline does for its reported forecasts (Section 6) produces a conditional forecast rather than a genuine one. A production-grade analogue would require either an iterative scoring process that feeds the model’s own forecasts back in as lag features, or the substitution of outdoor-weather forecasts (themselves imperfect) for their realised counterparts, either of which would be expected to result in larger errors than those reported in this paper.

7.6 Which model would be recommended for practical smart-home deployment?

Considering the accuracy, interpretability, uncertainty quantification, computational cost, and ease of deployment, XGBoost is the most optimal model. It has shown the best accuracy on all metrics. The model's features importances (gain, Figure 5) can be used for interpretability, providing a justification of the model's decisions that a homeowner would understand. XGBoost training and prediction is sufficiently fast and does not require the substantial infrastructure and resources training a transformer-based model would need, making it more deployment-friendly than Chronos. The only significant disadvantage is the lack of native support for uncertainty quantification. One could theoretically address this concern by training a quantile-regression model instead of a standard regression one. Overall, if sufficient historical data about a particular house is available for training, XGBoost would be the optimal choice due to its combination of high performance and relatively easy deployment. However, in the absence of such data, Chronos would be a better option due to its strong zero-shot performance.

8. Limitations and Future Work

There are several limitations to consider for these results. Firstly, the evaluation conducted on XGBoost is a "conditional" forecast for the covariates necessitating future information (Section 7.5), and a true iterative multi-step evaluation would exhibit a smaller, but potentially still positive, advantage over SARIMAX and Chronos. Secondly, all models systematically underestimate the largest demand peaks, which are likely due to ad-hoc appliance usage not captured by the available covariates; including appliance-level sub-metering or occupancy sensing, if available, would plausibly reduce this deficit. Thirdly, the Gaussian confidence intervals from SARIMAX, while capturing the true mean, exhibit physically implausible negative bounds, suggesting that a log-link or a different error-family would be more appropriate for this use-case. Fourthly, Chronos was evaluated completely zero-shot; a small amount of fine-tuning on the consumption pattern of this household, or an ensemble of Chronos and XGBoost, would be expected to close much of the performance-gap, as preliminary investigations indicate that fine-tuning foundation models can often bridge most of the performance-gap with task-specific models (Ansari et al., 2024). Lastly, the evaluation was performed on a single household, and it is unclear how these findings generalise to other households, seasonal variations, or the heating season not included in this analysis; a validation on a second household or an annual dataset would be informative.

9. Conclusion and Model Recommendation

Across benchmark, statistical, machine-learning and foundation-model approaches, a tuned XGBoost model using lag, rolling-window and time-of-day features showed the highest accuracy and best calibration (lowest bias) in forecasts of hourly appliance energy use, significantly outperforming both the best naive model and the zero-shot time-series foundation model. This said, the zero-shot foundation model's ability to compete without any task-specific training is also a promising observation, and reduces the practical gap between generic and tailored approaches to a certain extent. For the considered use-case, application domain and deployment scenario, XGBoost is then recommended on both accuracy, interpretability and deployment-cost grounds, with the caveat that the compared advantage was made in a conditional rather than full forecasting setting, and that all approaches showed shortcomings in capturing unscheduled demand increases which none of the considered covariates appeared sufficient to address.

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